

Frontend Interview Handbook — Part 4: SEO, Technical SEO, AEO & GEO

This is the section most directly tied to ViewEngine's actual product. Know this cold.

SECTION A: TECHNICAL SEO FUNDAMENTALS

Q1: What's the difference between on-page SEO, technical SEO, and off-page SEO?

Answer: **On-page SEO** is content and HTML-level factors you control per page — titles, headings, content quality, internal links. **Technical SEO** is the infrastructure layer — crawlability, indexability, site speed, rendering strategy, structured data, sitemaps, canonicalization. **Off-page SEO** is external signals — backlinks, brand mentions, citations from other sites. As a frontend engineer, your work sits almost entirely in technical SEO and the implementation side of on-page SEO — that's the intersection to speak to confidently.

Q2: What is a sitemap, and what's the frontend engineer's role in maintaining one?

Answer: A sitemap (usually `sitemap.xml`) is a machine-readable list of a site's URLs, helping crawlers discover pages efficiently — especially important for large sites or pages with few internal links pointing to them. For programmatic SEO (thousands of dynamically generated pages), the sitemap typically needs to be **generated dynamically** from the same data source that generates the pages, not hand-maintained, and split into multiple sitemap files if it exceeds the ~50,000 URL limit per file (using a sitemap index file to reference them).

```
tsx

// app/sitemap.ts (Next.js App Router built-in support)
export default async function sitemap() {
  const locations = await getAllLocations();
  return locations.map(loc => ({
    url: `https://example.com/locations/${loc.slug}`,
    lastModified: loc.updatedAt,
  }));
}
```

Q3: What is `robots.txt` for, and what are common mistakes made with it?

Answer: `robots.txt` tells crawlers which paths they're allowed/disallowed from crawling. It lives at the domain root (`/robots.txt`). Common mistakes: accidentally blocking the entire site (`Disallow: /`) left over from a staging environment pushed to production; blocking JS/CSS files needed to render the page, which prevents crawlers from properly

rendering and understanding content; confusing "disallow crawling" with "prevent indexing" — a disallowed page can still get indexed (e.g., via external links) with no description, since Google never fetched it to confirm what's there. To actually prevent indexing, use a `noindex` meta tag or header instead, and make sure the page isn't also blocked in robots.txt (blocking crawling prevents the crawler from ever seeing the noindex tag).

```
User-agent: *  
Disallow: /admin/  
Allow: /  
  
Sitemap: https://example.com/sitemap.xml
```

Q4: What is canonicalization, and why does it matter?

Answer: A canonical URL tells search engines which version of a page is the "authoritative" one when duplicate or near-duplicate content exists at multiple URLs (e.g., with/without trailing slash, with tracking query params, `www` vs non-`www`, paginated or filtered variants of a listing page). Without it, duplicate content can split ranking signals across URLs or cause the wrong version to be indexed.

```
html  
  
<link rel="canonical" href="https://example.com/products/widget" />
```

Follow-up: "You have a product page accessible at `/product/123` and `/product/123?ref=email` — what do you do?" A: Set the canonical tag on both to point to the clean `/product/123` URL, so ranking signals consolidate there regardless of how the page was accessed.

Q5: What is a 404 vs a "soft 404," and why is a soft 404 worse?

Answer: A real 404 returns HTTP status 404 (or 410 for permanently gone), telling crawlers unambiguously "this page doesn't exist, stop indexing it." A **soft 404** is when a page shows "not found" content to the user but actually returns a 200 OK status — common in SPAs that handle routing client-side and just render an error component without changing the server status code. Crawlers may still index these as valid pages with empty/error content, polluting the index. Frontend engineers building client-routed SPAs need to make sure not-found routes are either server-rendered with a true 404 status, or explicitly handled if using something like Next.js (`notFound()` in App Router triggers a proper 404 response).

Q6: How does URL structure affect SEO? What makes a "good" URL?

Answer: Clean, human-readable, keyword-relevant URLs help both users and crawlers understand page content at a glance, and tend to get better click-through in search results. Prefer `/blog/nextjs-seo-guide` over `/blog?id=48291`. Keep URLs stable — changing URL structure without proper 301 redirects loses accumulated ranking signal and breaks external links. For programmatic SEO, the URL pattern itself (e.g., `/locations/[city]/[service]`) often is the primary organizing structure for the whole content system.

SECTION B: STRUCTURED DATA & METADATA

Q7: What is JSON-LD structured data, and how would you implement it in a Next.js page?

Answer: JSON-LD is a script tag containing a JSON object describing the page's content in schema.org vocabulary — explicitly telling search engines and AI crawlers what type of entity is on the page (Article, Product, LocalBusiness, FAQPage, etc.) rather than leaving it to infer from text. This increases eligibility for rich results (star ratings, FAQ dropdowns in search) and gives AI answer engines an unambiguous, structured source to extract from.

tsx

```
function ProductSchema({ product }) {  
  const jsonLd = {  
    "@context": "https://schema.org",  
    "@type": "Product",  
    "name": product.name,  
    "description": product.description,  
    "offers": {  
      "@type": "Offer",  
      "price": product.price,  
      "priceCurrency": "USD",  
    },  
  };  
  return (  
    <script  
      type="application/ld+json"  
      dangerouslySetInnerHTML={{ __html: JSON.stringify(jsonLd) }}  
    />  
  );  
}
```

Common mistake: Adding structured data that doesn't match the visible content on the page (e.g., claiming a rating that isn't shown to users) — this violates search engine guidelines and can trigger manual penalties.

Q8: What Open Graph and Twitter Card tags matter, and why do they matter for AI/chat platforms too?

Answer:

```
html
```

```
<meta property="og:title" content="Page Title" />
<meta property="og:description" content="Concise description." />
<meta property="og:image" content="https://example.com/preview.png" />
<meta property="og:url" content="https://example.com/page" />
<meta name="twitter:card" content="summary_large_image" />
```

These control how a link preview renders when shared — in social platforms, Slack, iMessage, and increasingly when a URL is shared into or referenced by chat/AI platforms. A missing or generic OG image/description is a small but real trust/credibility signal.

Q9: What makes a title tag and meta description "good" from an SEO/AEO perspective?

Answer: Title: unique per page, front-load the primary keyword/topic naturally, keep to roughly 50-60 characters to avoid truncation in search results, avoid keyword stuffing. Meta description: roughly 150-160 characters, written as a genuinely useful summary a human would want to read — not stuffed with keywords, since modern engines (and AI systems evaluating trustworthiness) penalize or ignore unnatural text. For programmatic pages, template these dynamically from real data (not one static template reused thousands of times), since near-duplicate metadata at scale is a known red flag.

SECTION C: AEO (ANSWER ENGINE OPTIMIZATION) & GEO (GENERATIVE ENGINE OPTIMIZATION)

Q10: What is AEO, and how is it different from traditional SEO?

Answer: AEO (Answer Engine Optimization) is optimizing content to be selected as *the direct answer* by systems like Google's Featured Snippets, voice assistants, and AI Overviews — rather than just ranking as a blue link the user has to click through. This means structuring content to directly and concisely answer likely questions near the top of a page, often in Q&A format, with clear, extractable statements rather than answers buried in narrative prose.

Q11: What is GEO (Generative Engine Optimization), and how does it differ from AEO?

Answer: GEO is optimizing specifically for being cited, quoted, or recommended by generative AI systems (ChatGPT, Perplexity, Claude, Gemini) when they synthesize an answer from multiple sources — rather than a single engine returning one snippet. It overlaps heavily with AEO but has some distinct considerations: LLMs weigh source credibility/consistency across the web (being mentioned similarly across multiple independent sites reinforces the "fact" for the model), favor content that's clearly attributable and factual, and may draw from a broader retrieval index than what shows up in traditional top-10 search results.

Why this matters for the interview: Being able to articulate the AEO vs GEO distinction clearly — most people conflate them — is a strong signal you understand ViewEngine's actual positioning, not just generic SEO.

Q12: As a frontend engineer, what specific things can you build/implement that directly improve AEO/GEO outcomes?

Answer: Concrete, defensible list:

- Structuring content with clear heading hierarchy and direct-answer paragraphs near the top (not just for humans — for extraction).
 - Implementing FAQ-schema-marked Q&A sections where relevant.
 - Ensuring SSR/SSG so content is actually present in the HTML crawlers/AI retrieval fetch — the single highest-leverage frontend decision here.
 - Fast, stable pages (Core Web Vitals) — slow or shifting pages are deprioritized both by traditional ranking and by systems that time out on fetches.
 - Clean internal linking so crawlers/retrieval systems can discover the full breadth of a site's content, reinforcing topical authority.
 - Consistent structured data across a site so entities (products, locations, authors) are unambiguous.
-

Q13: How would you verify whether a page is actually being crawled/indexed and rendered correctly by a search engine?

Answer: Google Search Console's URL Inspection tool shows the last crawled version and lets you test live rendering. For AI-specific validation there's less official tooling, but practical checks: view the page's raw HTML (`curl` or "view source," not the rendered DOM) to confirm core content is actually present server-side, not injected only by client JS; check `robots.txt` isn't blocking it; verify structured data with Google's Rich Results Test.

Q14: What's the difference between crawl budget and index bloat, and why do they matter for a large programmatic SEO site?

Answer: **Crawl budget** is the finite amount of crawling attention a search engine allocates to a given site — on very large sites, if crawl budget is wasted on low-value pages (duplicate filter combinations, thin auto-generated pages, infinite faceted navigation), high-value pages may get crawled less often or not at all. **Index bloat** is having a large number of low-quality/thin/near-duplicate pages indexed, which can dilute a site's perceived quality overall. For programmatic SEO specifically, this means being deliberate: only generate pages with genuinely differentiated, useful content, use `(noindex)` or canonicalization for thin variants, and prioritize which pages get pre-rendered/most frequently crawled.

Rapid-Fire Round

- **Q: What's a 301 vs 302 redirect?** A: 301 = permanent, passes ranking signal to the new URL; 302 = temporary, search engines generally keep the original URL indexed. Using 302 for a permanent change is a common mistake that loses SEO equity.
 - **Q: What's "thin content"?** A: Pages with little unique or useful value — auto-generated pages with minimal differentiation are a common trap in programmatic SEO if not done carefully.
 - **Q: What's E-E-A-T?** A: Experience, Expertise, Authoritativeness, Trustworthiness — a framework Google's quality raters (and increasingly AI systems evaluating source credibility) use to judge content quality, especially for topics affecting health/finance/safety.
 - **Q: What's a breadcrumb, and why does it help SEO?** A: A navigational trail (Home > Category > Product) shown to users and often marked up with BreadcrumbList schema — helps both users and crawlers understand site hierarchy, and often renders directly in search results.
 - **Q: Why does page speed matter for SEO beyond Core Web Vitals as a ranking factor?** A: Slow pages increase bounce rate and reduce how many pages a crawler can fetch within its allotted crawl budget/time per site.
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End of Part 4. Say "next part" for Part 5: Performance, Core Web Vitals & Browser Internals.